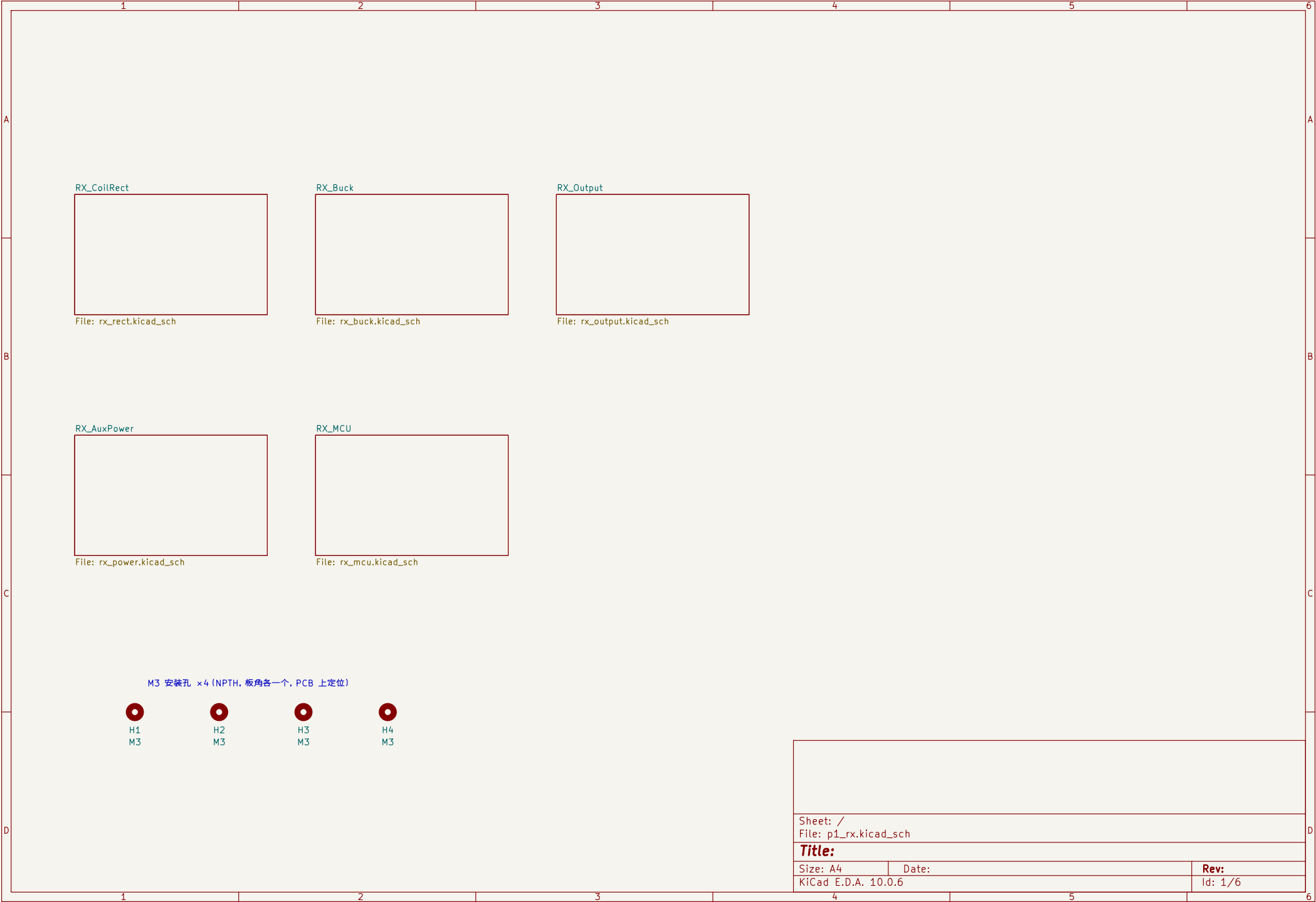
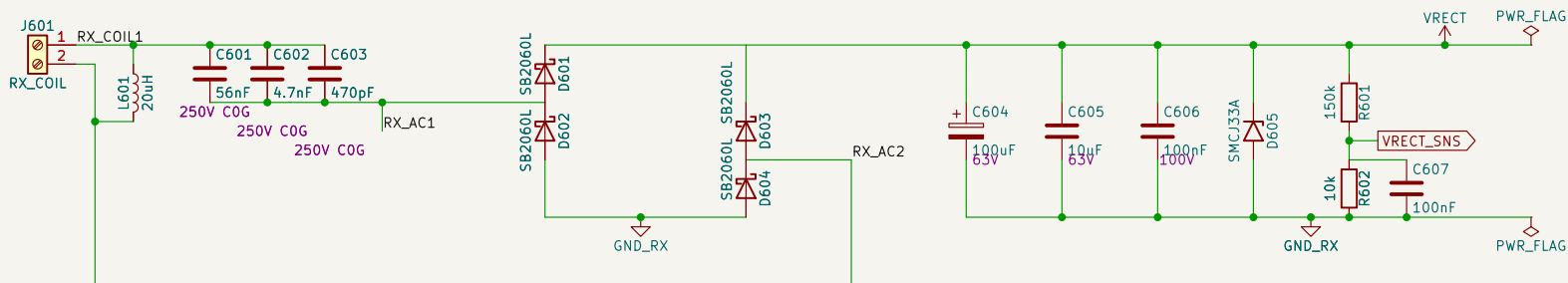






RX coil Lr, off-board via J601, model only

RX side, isolated ground GND_RX. Series compensation: $C_r = 1/(w^2 L_r) = 61.1 \text{ nF}$ at 144 kHz $\rightarrow C601 || C602 || C603 = 61.17 \text{ nF}$ (COG/film).

Ideal-model estimate (Vrect 30 V, 120 W at the cap, Buck 95 %): Irect = 4.21 A, $I_{Lr} = 1.11 \times \text{Irect} = 4.7 \text{ Arms}$, $V_{Cr} = 85 \text{ Vrms}$.

Mutual inductance $M = 0.9 \text{ Vrect} / (w I_{Lr}) = 7.5 \text{ uH}$, $k = 0.37$ for $L_t = L_r = 20 \text{ uH}$ (design point).

D601-D604 SB2060L Schottky (20 A / 60 V): bridge conduction loss about $2 \times 0.5 \text{ V} \times 4.21 \text{ A} = 4.2 \text{ W}$.

D605 SMCJ33A: VRWM 33 V above the 30 V nominal VRECT; clamp about 53 V stays under the 60 V rectifiers and 100 V MOSFETs. $V_{RECT_SNS} = V_{RECT} \times 10k / 160k$.

Sheet: /RX_CoilRect/
File: rx_rect.kicad_sch

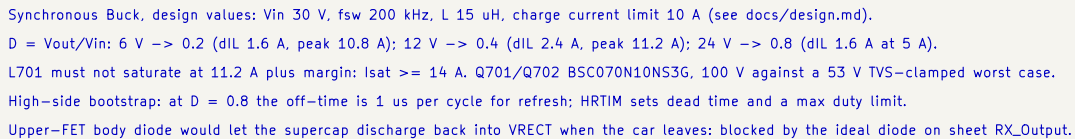
Title:

Size: A4 Date:

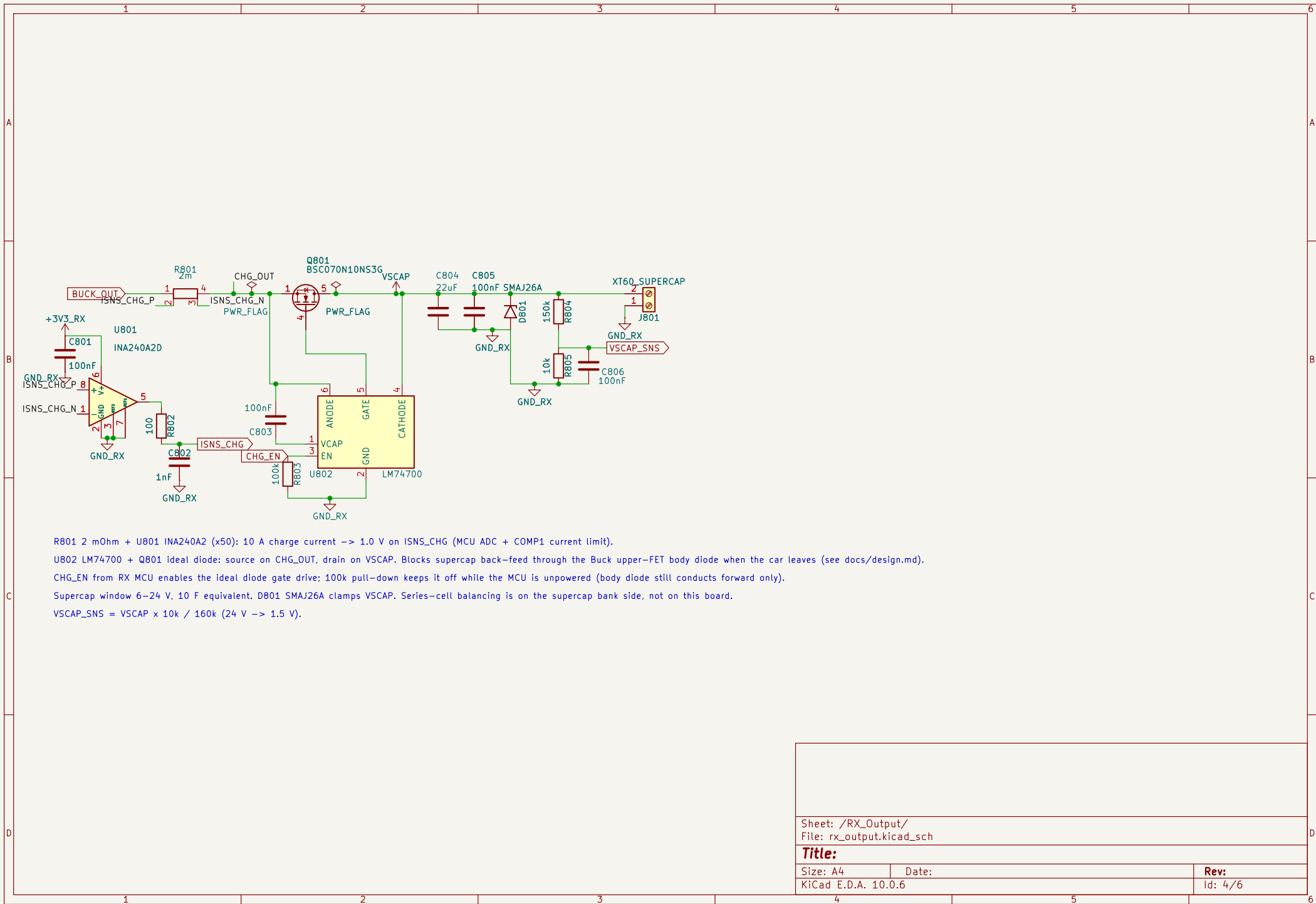
KiCad E.D.A. 10.0.6

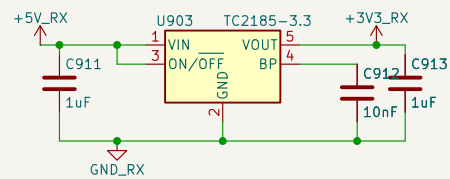
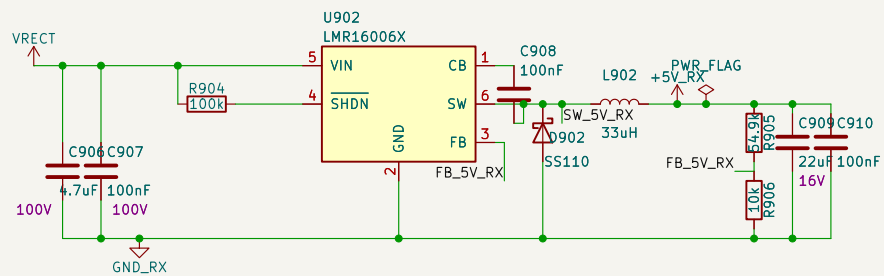
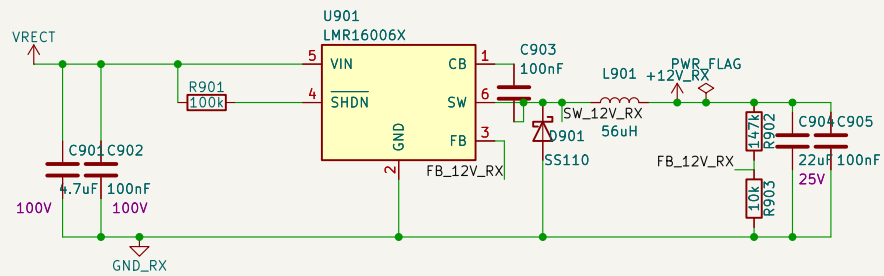
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RX aux power, powered from VRECT only: when the car leaves, the RX MCU and Buck lose power and the ideal diode turns off.

U901: $0.765 \text{ V} \times (1 + 147\text{k}/10\text{k}) = 12.0 \text{ V}$ for UCC27211 VDD. U902: $0.765 \text{ V} \times (1 + 54.9\text{k}/10\text{k}) = 4.96 \text{ V}$. U903 TC2185-3.3: MCU + INA240 only (150 mA max).

Input parts rated 100 V: VRECT is 30 V nominal, about 53 V worst case at the SMCJ33A clamp (LMR16006 abs max 60 V).

Sheet: /RX_AuxPower/
File: rx_power.kicad_sch

Title:

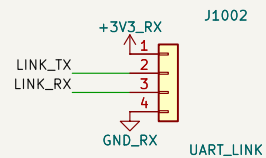
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Id: 5/6

ADC_VRef+ must not float: tied to +3V3_RX through FB1001 with 1 uF + 100 nF.



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